Supplementary Material: Detailed Mathematical Formulations of Coordinated Dispatch

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I. Nomenclature

Symbol	Definition
$\mathcal{T}, \Delta t$	Dispatch intervals $\{1, \dots, T\}$ and interval duration.
$\mathcal{N}, \mathcal{E}, r$	DN buses, branches directed away from the root, and root bus.
$\mathcal{M}, \mathcal{M}_n$	MGs and MGs connected to bus n.
$\mathcal{G}^{\mathrm{DN}}, \mathcal{G}_n^{\mathrm{DN}}$	DN generators and generators at bus n.
$\mathcal{R}_m, \mathcal{G}_m, \mathcal{B}_m$	PV units a, generators g, and storage units b in MG m.
$P_{ij,t}, Q_{ij,t}$	Sending-end active/reactive branch powers.
$v_{n,t}, \ell_{ij,t}$	Squared voltage/current magnitudes.
P_t^0, Q_t^0	Upstream active/reactive imports; negative values denote exports.
$g_{g,t}^P, g_{g,t}^Q$	DN generator active/reactive outputs.
$p_{m,t}(\boldsymbol{x}_m), q_{m,t}$	MG-side and DN-side PCC powers; positive for MG imports.
$p_{m,a,t}^{\mathrm{PV}}, p_{m,g,t}^G$	Utilized MG PV output and generator output.
$p_{m,b,t}^{\mathrm{ch}}, p_{m,b,t}^{\mathrm{dis}}$	Storage charging/discharging powers.
$e_{m,b,t}, e_{m,b}^{\mathrm{ini}}$	Stored energy and prescribed initial energy.
$d_{n,t}^P, d_{n,t}^Q$	Background DN demands, excluding MG exchanges.
$L_{m,t}, \overline{p}_{m,a,t}^{\mathrm{PV}}$	MG load and available PV power.
r_{ij}, x_{ij}	Branch resistance/reactance.
$\lambda_t^{\mathrm{up}}, c_{\mathrm{loss}}$	Upstream energy and network-loss cost coefficients.
$a_g^{\mathrm{DN}}, b_g^{\mathrm{DN}}, c_g^{\mathrm{DN}}$	Quadratic, linear, and constant DN generation cost coefficients.
$a_{m,g}^{\mathrm{MG}}, b_{m,g}^{\mathrm{MG}}$	MG quadratic/linear generation cost coefficients.
$c_{m,b}^{\mathrm{ch}}, c_{m,b}^{\mathrm{dis}}, c_{m,a}^{\mathrm{curt}}$	Storage charging, discharging, and PV curtailment cost coefficients.
$\eta_{m,b}^{\mathrm{ch}}, \eta_{m,b}^{\mathrm{dis}}$	Charging/discharging efficiencies.
$R_g^{\uparrow}, R_g^{\downarrow}$	DN generator ramp limits per interval.
$R_{m,g}^{\mathrm{MG}}$	MG generator ramp limit per interval.
$\underline{V}_n, \overline{V}_n, \overline{I}_{ij}$	Voltage and current magnitude limits.
$\underline{(\cdot)}, \overline{(\cdot)}$	Lower/upper operating bounds.
$\delta_{n,r}$	One if $n = r$; zero otherwise.
$\boldsymbol{x}_m$	Internal decision vector of MG m.
$\boldsymbol{y}$	DN decision vector.
$\boldsymbol{p}_m(\boldsymbol{x}_m) \in \mathbb{R}^T$	Affine PCC mapping: $\mathrm{col}_{t\in\mathcal{T}}(p_{m,t}(\boldsymbol{x}_m))$.
$\boldsymbol{q}_m \in \mathbb{R}^T, \boldsymbol{q}$	DN-side PCC exchange vector of MG m and its stack over all MGs.
$\mathcal{X}_m$	Convex feasible set of the internal MG decisions $\boldsymbol{x}_m$.
$\mathcal{Y}$	Convex DN feasible set for $(\boldsymbol{y}, \boldsymbol{q})$.
$C^{\mathrm{DN}}(\boldsymbol{y}), C_m(\boldsymbol{x}_m)$	DN and MG convex operating-cost functions.
$\boldsymbol{\omega}_m \in \mathbb{R}^T$	Dual multiplier of $\boldsymbol{p}_m(\boldsymbol{x}_m) - \boldsymbol{q}_m = \boldsymbol{0}$.

II. Objective Function

$$\min_{\boldsymbol{y},\boldsymbol{q},\{\boldsymbol{x}_m\}} \quad C^{\mathrm{DN}}(\boldsymbol{y}) + \sum_{m\in\mathcal{M}} C_m(\boldsymbol{x}_m), \tag{1a}$$

$$\text{s.t.} \quad (\boldsymbol{y}, \boldsymbol{q}) \in \mathcal{Y}, \tag{1b}$$

$$\boldsymbol{x}_m \in \mathcal{X}_m, \quad \forall m \in \mathcal{M}, \tag{1c}$$

$$\boldsymbol{p}_m(\boldsymbol{x}_m) - \boldsymbol{q}_m = \boldsymbol{0}, \quad \forall m \in \mathcal{M} \quad (\boldsymbol{\omega}_m). \tag{1d}$$

The DN and MG operating costs are

$$C^{\mathrm{DN}}(\boldsymbol{y}) = \Delta t \sum_{t\in\mathcal{T}} \left[\lambda_t^{\mathrm{up}} P_t^0 + c_{\mathrm{loss}} \sum_{(i,j)\in\mathcal{E}} r_{ij}\ell_{ij,t} + \sum_{g\in\mathcal{G}^{\mathrm{DN}}} \left(a_g^{\mathrm{DN}} (g_{g,t}^P)^2 + b_g^{\mathrm{DN}} g_{g,t}^P + c_g^{\mathrm{DN}} \right) \right], \tag{2a}$$

$$C_m(\boldsymbol{x}_m) = \Delta t \sum_{t\in\mathcal{T}} \left\{ \sum_{g\in\mathcal{G}_m} \left[a_{m,g}^{\mathrm{MG}} (p_{m,g,t}^G)^2 + b_{m,g}^{\mathrm{MG}} p_{m,g,t}^G \right] + \sum_{b\in\mathcal{B}_m} \left(c_{m,b}^{\mathrm{ch}} p_{m,b,t}^{\mathrm{ch}} + c_{m,b}^{\mathrm{dis}} p_{m,b,t}^{\mathrm{dis}} \right) + \sum_{a\in\mathcal{R}_m} c_{m,a}^{\mathrm{curt}} \left(\overline{p}_{m,a,t}^{\mathrm{PV}} - p_{m,a,t}^{\mathrm{PV}} \right) \right\}. \tag{2b}$$

III. Constraints

A. DN Feasible Set

The set $\mathcal{Y}$ is defined by (3)–(6).

Nodal power balance:

$$\delta_{n,r}P_t^0 + \sum_{i:(i,n)\in\mathcal{E}} (P_{in,t} - r_{in}\ell_{in,t}) + \sum_{g\in\mathcal{G}_n^{\mathrm{DN}}} g_{g,t}^P - \sum_{j:(n,j)\in\mathcal{E}} P_{nj,t} = d_{n,t}^P + \sum_{m\in\mathcal{M}_n} q_{m,t}, \tag{3a}$$

$$\delta_{n,r}Q_t^0 + \sum_{i:(i,n)\in\mathcal{E}} (Q_{in,t} - x_{in}\ell_{in,t}) + \sum_{g\in\mathcal{G}_n^{\mathrm{DN}}} g_{g,t}^Q - \sum_{j:(n,j)\in\mathcal{E}} Q_{nj,t} = d_{n,t}^Q. \tag{3b}$$

Branch-flow relations and SOC relaxation:

$$v_{j,t} = v_{i,t} - 2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) + (r_{ij}^2 + x_{ij}^2)\ell_{ij,t}, \tag{4a}$$

$$\left\| \left(2P_{ij,t},\, 2Q_{ij,t},\, v_{i,t} - \ell_{ij,t}\right)^{\mathsf{T}} \right\|_2 \le v_{i,t} + \ell_{ij,t}. \tag{4b}$$

Network operating limits:

$$\underline{V}_n^{\,2} \le v_{n,t} \le \overline{V}_n^{\,2}, \quad v_{r,t} = 1, \tag{5a}$$

$$0 \le \ell_{ij,t} \le \overline{I}_{ij}^{\,2}, \tag{5b}$$

$$-\overline{P}_{ij} \le P_{ij,t} \le \overline{P}_{ij}, \tag{5c}$$

$$-\overline{Q}_{ij} \le Q_{ij,t} \le \overline{Q}_{ij}, \tag{5d}$$

$$-\overline{P}^0 \le P_t^0 \le \overline{P}^0, \tag{5e}$$

$$-\overline{Q}^0 \le Q_t^0 \le \overline{Q}^0. \tag{5f}$$

DN generator limits:

$$\underline{g}_g^P \le g_{g,t}^P \le \overline{g}_g^P, \tag{6a}$$

$$\underline{g}_g^Q \le g_{g,t}^Q \le \overline{g}_g^Q, \tag{6b}$$

$$-R_g^{\downarrow} \le g_{g,t}^P - g_{g,t-1}^P \le R_g^{\uparrow}, \quad t = 2,\dots,T. \tag{6c}$$

B. MG Feasible Sets and PCC Mapping

For each $m \in \mathcal{M}$, the internal power balance defines the affine PCC mapping by

$$p_{m,t}(\boldsymbol{x}_m) = L_{m,t} - \sum_{a\in\mathcal{R}_m} p_{m,a,t}^{\mathrm{PV}} - \sum_{g\in\mathcal{G}_m} p_{m,g,t}^G - \sum_{b\in\mathcal{B}_m} (p_{m,b,t}^{\mathrm{dis}} - p_{m,b,t}^{\mathrm{ch}}). \tag{7}$$

The set $\mathcal{X}_m$ is defined by (8)–(10).

PV and dispatchable generation:

$$0 \le p_{m,a,t}^{\mathrm{PV}} \le \overline{p}_{m,a,t}^{\mathrm{PV}}, \tag{8a}$$

$$0 \le p_{m,g,t}^G \le \overline{p}_{m,g}^G, \tag{8b}$$

$$-R_{m,g}^{\mathrm{MG}} \le p_{m,g,t}^G - p_{m,g,t-1}^G \le R_{m,g}^{\mathrm{MG}}, \quad t = 2,\dots,T. \tag{8c}$$

Energy storage:

$$0 \le p_{m,b,t}^{\mathrm{ch}} \le \overline{p}_{m,b}^{\mathrm{ch}}, \tag{9a}$$

$$0 \le p_{m,b,t}^{\mathrm{dis}} \le \overline{p}_{m,b}^{\mathrm{dis}}, \tag{9b}$$

$$e_{m,b,t} = e_{m,b,t-1} + \Delta t \left(\eta_{m,b}^{\mathrm{ch}} p_{m,b,t}^{\mathrm{ch}} - \frac{p_{m,b,t}^{\mathrm{dis}}}{\eta_{m,b}^{\mathrm{dis}}} \right), \tag{9c}$$

$$\underline{e}_{m,b} \le e_{m,b,t} \le \overline{e}_{m,b}, \quad t = 0,\dots,T, \tag{9d}$$

$$e_{m,b,0} = e_{m,b}^{\mathrm{ini}}. \tag{9e}$$

PCC exchange limits:

$$\underline{p}_m \le p_{m,t}(\boldsymbol{x}_m) \le \overline{p}_m. \tag{10}$$

C. Variable Collections

The three decision blocks are

$$\boldsymbol{x}_m = \mathrm{col}\big(\boldsymbol{p}_m^{\mathrm{PV}}, \boldsymbol{p}_m^{G}, \boldsymbol{p}_m^{\mathrm{ch}}, \boldsymbol{p}_m^{\mathrm{dis}}, \boldsymbol{e}_m\big), \quad m \in \mathcal{M}, \tag{11a}$$

$$\boldsymbol{y} = \mathrm{col}\big(\boldsymbol{P}, \boldsymbol{Q}, \boldsymbol{v}, \boldsymbol{\ell}, \boldsymbol{P}^0, \boldsymbol{Q}^0, \boldsymbol{g}^P, \boldsymbol{g}^Q\big), \tag{11b}$$

$$\boldsymbol{q}_m = \mathrm{col}_{t\in\mathcal{T}}(q_{m,t}), \quad \boldsymbol{q} = \mathrm{col}_{m\in\mathcal{M}}(\boldsymbol{q}_m). \tag{11c}$$

Thus, $\boldsymbol{x}_m$ collects MG generation, storage charging/discharging, and stored energy; $\boldsymbol{y}$ collects DN branch flows, squared voltages/currents, upstream exchanges, and DN generator outputs; $\boldsymbol{q}$ collects the DN-side PCC exchange profiles. Each block stacks its scalar variables over the relevant units, buses, or branches and all dispatch periods. Energy blocks include $t = 0$. The MG PCC profile $\boldsymbol{p}_m(\boldsymbol{x}_m)$ is determined by (7).